

Iris Flower Classification Project Report

1. Introduction

The Iris Flower Classification project was developed as part of the CodeAlpha Data Science Internship Program. The purpose of this project is to classify Iris flowers into different species using Machine Learning algorithms.

The Iris dataset is one of the most widely used datasets for classification tasks and contains measurements of flower characteristics that can be used to identify flower species.

2. Objective

The main objectives of this project are:

- To understand classification problems in machine learning.
 - To analyze and visualize the Iris dataset.
 - To build a predictive classification model.
 - To evaluate model performance using standard metrics.
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3. Dataset Description

The dataset used in this project is **Iris.csv**.

Features

- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm

Target Variable

Species:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Dataset Statistics:

- Total Records: 150
- Total Features: 4

- Total Classes: 3

The dataset contains no missing values and is balanced across all species.

4. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
 - Checked dataset dimensions and structure.
 - Verified missing values.
 - Separated features and target variables.
 - Prepared data for model training.
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5. Exploratory Data Analysis

EDA was performed to understand patterns within the dataset.

Analyses Performed

- Species Distribution Analysis
- Pair Plot Visualization
- Correlation Heatmap
- Statistical Summary

These visualizations helped identify relationships among the flower measurements and species classes.

6. Model Development

Algorithm Used

Random Forest Classifier

Training Process

- Train-Test Split: 80:20
- Random State: 42
- Number of Trees: 100

The model was trained using the training dataset and tested on unseen samples.

7. Model Evaluation

The following metrics were used:

- Accuracy Score
- Confusion Matrix
- Classification Report

Results

Accuracy Achieved:

100%

The model correctly classified all test samples and demonstrated excellent predictive performance.

8. Feature Importance Analysis

Feature importance analysis was conducted to determine which flower measurements contributed most to classification performance.

Petal-related measurements showed higher importance compared to sepal measurements.

9. Conclusion

The Iris Flower Classification project successfully demonstrated the use of Machine Learning for multiclass classification.

Using the Random Forest Classifier, the model achieved 100% accuracy on the testing dataset. The project highlights the effectiveness of data preprocessing, visualization, feature engineering, and machine learning algorithms in solving classification problems.

This project serves as a strong foundation for understanding supervised learning and classification techniques.

10. Tools and Libraries Used

- Python
- Pandas

- NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - Jupyter Notebook
 - Visual Studio Code
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Author

Vineet Singh

CodeAlpha Data Science Internship